

Love of Variety, Habit Formation, and Dynamic Advertising

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Roadmap

- 1 Motivation
- 2 Model
- 3 Simulation Design
- 4 Results: Consumer Behavior
- 5 Next Steps

Motivation: Data Privacy, Ad Targeting, and Consumer Agency

- Amazon anti-trust scrutiny
 - Amazon accounts for almost 40% of US e-commerce sales (Statista, 2023)
 - Able to use consumer data to target ads and build market share as both retailer and platform
- Consumer agency over data (opt out)
 - Apple iOS 14.5 update: 85% opt-out rate upon rollout (Flurry Analytics, 2022)
 - Facebook ad revenue down \$13bn in 2022 due to app tracking transparency (ATT) policy

Key Concern

How do data privacy policies that limit firms' ability to target ads based on consumer data impact consumer welfare and firms' incentives to invest in product quality and advertising?

Why Does Variety Matter?

- Understanding the role of variety-seeking is valuable for modeling consumer behavior and evaluating the impact data privacy policies on both consumers and firms
- Canonical logit models ignore consumer preferences for variety, which are empirically important
- These preferences may shape consumer switching behavior, which in turn shapes firms' incentives to strategically target ads and promotions (i.e. to encourage switch to a higher markup product)

Research Question

How do consumer preferences for variety and habit impact the firms' targeting strategy and what are the welfare implications of data privacy policies that limit ad targeting?

What This Paper Does (So Far)

- ① **Model:** Model a discrete-choice demand system where the utility of a product depends on both deviation from the consumer's choice history θ_{it} and product quality X_{jt}
- ② **Simulation:** Characterize consumer and firm behavior across preference regimes (β_i, γ_i) and advertising specifications
- ③ **Estimation (coming):** Bring model to Nielsen scanner data to recover structural parameters
- ④ **Counterfactuals (coming):** Study advertising targeting policy and welfare implications

Today's talk: Model + simulation results

Literature Connections

Demand side

- Lattin (1987): State dependence in consumer choice
- Chintagunta (1999): Extension to panel data
- Dubé, Hitsch, Rossi (2010): Habit formation and switching costs

Advertising / firm side

- Shaffer and Zhang (1995): Theoretical basis for ad targeting in a two-period model
- Erdem & Keane (1996): Learning dynamics
- Zhang, Choi, Cheng (2025): Updated targeting game

Contribution

Unite the demand and supply side literatures by modeling consumer preferences for variety and habit formation in a dynamic discrete choice framework with strategic advertising.

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Consumer's Problem

Consumer i chooses product j in period t to maximize:

$$U_{ijt} = \underbrace{\beta_{ij} X_{jt}}_{\text{characteristic component}} + \underbrace{\gamma_i \log(1 + \Sigma_{ijt}^2)}_{\text{novelty component}} - \underbrace{\alpha p_{jt}(1 - a_{jt})}_{\text{ad-adjusted price}} + \xi_j + \varepsilon_{ijt}$$

- where $\Sigma_{ijt} = X_{jt} - \theta_{it}$ is the deviation of product j 's characteristic from consumer i 's previously chosen characteristics θ_{it} in period t .
- at the moment, $\theta_{it} = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$, or the mean of previously chosen product characteristics.

Key parameters:

- $\beta_i > 0$: marginal utility of quality
- $\gamma_i > 0$: variety-seeking
- $\kappa > 0$: advertising shifter

Choice probability (MNL):

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}}$$

Follows from $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$

State Variable: History of Choices

History of Choices θ_{it}

$$\theta_{it} = f(\{X_{js}\}_{s < t}) = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$$

The consumer's history is summarized by the mean of previously chosen product characteristics.

Initialization strategy:

- 1 **Initial state:** Simulate $T_{\text{prior}} = 5$ pre-sample periods from $J \in [1, 2, 3, 4, 5]$ to generate an experience-based prior.

Initial history yields more heterogeneous behavior — more on this shortly.

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Simulation Parameters

Table: Parameters

Param.	Value	Description
T	100	Periods
J	5	Products
S	1,000	Simulations
α	0	Price (off)
ξ_j	0	Quality (off)
β_{df}	0	Discount factor (off)
β_{ij}	2	Quality sensitivity
γ_i	{6, 9, 12}	Variety-seeking

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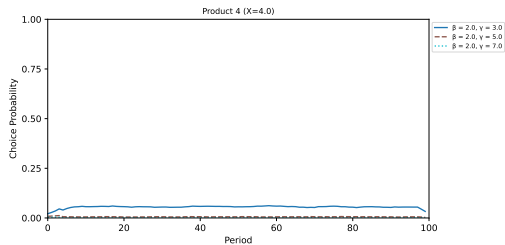
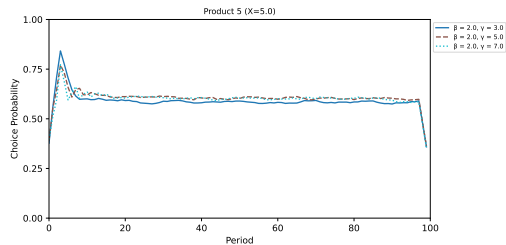
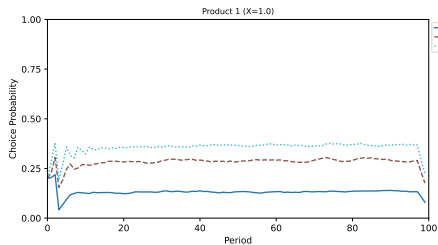
Naive Measurement

- Near static choice probabilities
- Any variation due to ε_{ijt} , not consumer preferences
- CCPs are not informative about consumer behavior or firm incentives
- Nearly all mass on Product 5, which has the highest quality X_{jt}

Table: Naive CCPs

Product	CCP
Product 1	0.02%
Product 2	0.54%
Product 3	4.05%
Product 4	22.3%
Product 5	72.0%

LOV Specification



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Road to Estimation

① Multiple consumers

Extend simulation to $I = 5$ consumers who have heterogeneous (β_i, γ_i) ; aggregate to market shares

② Consumer value function

Move from reduced-form simulation to solving the full dynamic discrete choice with continuation values

③ Nielsen scanner data

Match simulated moments to data; identify (β_i, γ_i) using BLP instruments

④ Firm's Problem

Ad targeting policy; profit maximization; counterfactuals

Summary

- ① **Model:** Consumer utility depends on deviation from experiences; nests variety-seeking and habit formation
- ② **History:** Rich behavioral dynamics; regime differences generate cross-sectional sorting
- ③ **Next:** Multiple consumers, structural estimation, Nielsen data

Thank you — questions welcome

Appendix

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Simulation Algorithm (Consumer)

- ➊ Draw $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$ for all i, j, t
- ➋ Initialize $\bar{X}_{i0}$ (set mean or prior mean)
- ➌ For $t = 1, \dots, T$:
 - ➊ Compute $U_{ijt} = \beta_i X_{jt} - \gamma_i (X_{jt} - \bar{X}_{it})^2 + \kappa a_{jt} + \varepsilon_{ijt}$
 - ➋ Set $j^* = \arg \max_j U_{ijt}$
 - ➌ Update $\bar{X}_{it+1} = \frac{1}{t} \sum_{k \leq t} x_{ijk^*}$
- ➍ Compute CCPs via log-sum-exp for numerical stability:
 $\log P_{ijt} = U_{ijt} - \log \sum_k e^{U_{ikt}}$
- ➎ Average over $S = 1,000$ simulation draws